



**QUICK USER GUIDE FOR THE COMPILATION OF THE ESA PRICE
BREAKDOWN FORMS**

Document Type	MAN - Manual / User Guide / Handbook
Reference	ESA-CIC-IAB-DRD-2026-6630-PP
Issue/Revision	01
Date of Issue	15/03/2026



Table of Contents

1. Background..... 3

2. PBF logic 3

3. Price conditions..... 4

4. Labour cost..... 5

5. Internal Special Facilities cost..... 7

6. Other Direct Costs 7

7. Total Company Costs 9

8. Total Company Price 10

9. Subcontractors..... 12

10. Management Reserve..... 13

11. PBF-2..... 13

12. PBF-3..... 14

13. SUPPORT 15

14. Note regarding the PSS-A forms..... 15

1. BACKGROUND

In ESA's Tender Actions and more in general in procurements made within ESA Programmes, any price quotation, irrespective of the price type, must be justified by bidders and detailed by using standard Price Breakdown Forms (previously PSS-A forms), in short 'PBF'.

The PBF are price reports produced with the ECOS software. The use of ECOS is mandatory for large projects (i.e. ESA Tender Actions with an overall price above EUR 10M across all contractual phases). For Tender Actions of lower value, the use of ECOS is an optional choice, unless differently indicated in the specific Conditions to Tender or price requirements.

Even when the use of ECOS is not mandatory, the ESA Industrial Audit Office and the ESA Cost Engineering Section highly recommend the use of ECOS for the preparation of a compliant price proposal.

In all cases where the bidder decides not to use ECOS, the price breakdown presentation shall respect the format of the PBF. For this reason, a template of the PBF is available as MS Excel file and published in esa-star. Tailoring or modification of the PBF is not allowed and may result in the down-marking or the rejection of the offer.

Bidders bear the responsibility to ensure a correct calculation of the price.

Where ECOS is not mandatory, the Tender Action Data Package shall include the required details to be presented in PBF. Typically, the requirement of:

- PBF1 (PSS-A2) is always mandatory; Annexes to PBF1 may be decided by the TEB.
- PBF2 (PSS-A8) may be decided by the TEB.
- PBF3 (PSS-A10 and 15) may be decided by the TEB.

Any other pre-defined price reports or PBF can only be printed using ECOS.

2. PBF LOGIC

The three main PBF (i.e. PBF-1, PBF-2, PBF-3), including all annexes, provide ESA with full visibility of the quoted price:

- PBF-1 shows the total price, broken down by price elements.

- PBF-2 shows the price of each Work-Package, broken down by price elements.
- PBF-3 shows the price distribution per period (normally by month), broken down by price elements.

The annexes provide cost details which appear consolidated in the main PBF tables.

The standard price elements in ESA contracts are the following:

- Labour cost
- Internal Special Facilities cost
- Other Direct Costs (Materials, External Services, Miscellaneous, Travel)
- Profit fee
- Cost Without Additional Charges
- Company Contributions

These elements are retained in all the main PBF.

3. PRICE CONDITIONS

The price conditions are normally part of the **costing requirements** published by ESA, or by the issuing entity in cases on non-ESA Tender Actions.

Price type and economic conditions

The **price type** refers to the definitions included in the ESA/REG/002 Annex II.

In case of *Firm and Fixed Price* (FFP) and *Ceiling Price to be converted to Firm and Fixed Price* (CP->FFP), the price shall cover the inflation of prices over the planned full duration of the contract. The term 'Firm' means that no adjustment for price inflation can be done after the signature of the contract. The term 'Fixed' means that no adjustment of the price can be done as result of the actual cost incurred.

In case of *Fixed Price with price Variation* (FPV) the cost shall respect the required **economic conditions** (i.e. the overall state of economy in the Country of the bidder at a given year and month; e.g. 07/2025) and therefore the financial inflation shall not be included. This type of price represents a risk sharing scheme on financial inflation: a formula based on inflation indices is agreed in advance and used to compensate the general inflation of prices.

It shall be remarked that the aim of the Price Variation Formula is always to compensate the Fixed Price for the general inflation experienced in the Country of the bidder, therefore it does not include any specific element of evolution of the bidder's cost structure.

Currency

Bidders are required to calculate the price using the *National Currency* (NC) of their Country of registration (unless the entity accounts are recorded using a different currency).

The price of ESA contracts is normally agreed in the EURO currency, bidders in Countries non adopting the EURO must propose a currency exchange rate.

- For FFP, the exchange rate used should be one representative of the period for which the work will be performed. When proposing a reasonable exchange rate, companies should refer to the European Central Bank publications. An analysis based on the duration of the contract may consider an average rate based on spot rates published in the past 6-months or 1-year and considering trends for extrapolation of possible future evolutions. A reasonable margin based on the volatility of the rate may be considered acceptable to protect the contractor financial exposure and to limit cost of hedging.
- For FPV, the exchange rate used must be the one published on the ESA Price Statistics Bulletin for the month and year of the required economic conditions. The reason for this requirement is linked to the use of EURO indices in the Price Variation Formula; these special indices produced for ESA include the effect of the exchange rate variation and assume the values published (which refer to the European Central Bank).

4. LABOUR COST

Labour cost refers to the cost of direct personnel to execute the contractual tasks.

The PBF requires that direct personnel is grouped by 'labour categories' or cost centres, and that an average cost rate for each category of labour is charged to the number of hours quoted to perform the work.

The cost rate shall be comprehensive of direct costs (i.e. salaries and social charges), as well as any direct (e.g. other personnel costs) and indirect (e.g. General & Administration) overheads; this combination is referred to as 'gross rate'.

The allowable cost is described within the Annex I of the ESA/REG/002.

The bidder shall quote

- the number of hours necessary to perform the work;
- the gross labour hourly rates;
- the average productive hours of each labour category.

Productive hours

Bidders shall refer to the ESA definition of *productive hours*: the normal number of hours that direct personnel record to funded and partially funded activities, including any self-funded R&D.

The word 'normal' refers to the requirement that the rates and overheads allocated to each unit of production shall not be significantly influenced as a consequence of a particular economic situation, abnormal production, uncommon practices or idle facilities/personnel.

In essence, for the calculation of the productive hours, the bidder shall consider an optimal production level.

The following tasks are excluded from productive hours : Proposals and Tenders, Training, internal general meetings, administrative, management and other non-productive time.

It is expected in general that the average productive hours are within 75-95% of the average worked hours.

Worked hours are related to the personnel work conditions and exclude any sort of absence.

ESA Audited entities

In case of entities audited by ESA for the agreement of rates and overheads, these entities shall always quote their prices using the agreed rates and provide the reference number of the agreement.

When a rates agreement has expired, the latest agreed rates shall be used as a reference to propose provisional escalated rates.

Entities never audited by ESA can propose reasonable rates in compliance to the ESA/REG/002 Annex I. ESA will assess the reasonability of the proposed rates and may enter into negotiation in those cases where values are found unacceptable.

Upon ESA request, the calculation of the proposed rates shall be provided in full detail.

5. INTERNAL SPECIAL FACILITIES COST

Internal Special Facilities (ISF) refer to special cost centres; typically, expensive assets engaged only in certain projects and at different usage, for example Thermal Vacuum Chambers, Clean Rooms, Manufacturing tools or workstations, software tools etc.

Isolating the cost of these cost centres in separate cost rates serves to improve the project cost estimating and provides customers with a fair price.

The bidders shall define the cost to be charged via the ISF rates, which may include the depreciation of the acquisition cost, maintenance, set-up and even operations, including the cost of dedicated operators where relevant. Possibly a portion of the company G&A expenditures may be added to the charged cost rates.

The unit of charge depends on the operation of each ISF. For example, a Clean Room may be charged by sqm/day; a manufacturing workstation by operations hours.

The cost allocated to ISF rates shall be excluded from the company overheads charged on Labour or procurements.

Audited and non-audited entities

The ISF rates quoted by audited and non-audited entities shall respect the same requirements as for Labour rates.

6. OTHER DIRECT COSTS

Other Direct Costs (ODC) include all those external costs necessary for the execution of the contract. These costs must be excluded from the Labour and ISF rates.

Include: Material, External Services, Miscellaneous and Travel.

Detail of each quoted expense shall be provided using the annexes (A-D) as required.

Purchases quoted under ODC become property of ESA and shall be delivered to ESA, unless differently indicated in ESA regulations or contractual agreements.

Where purchased goods are only partially used for the contract quoted, then only the relevant portion of their costs can be quoted.

Where calculated within the entity's rates structure, a *procurement overhead* may be quoted in addition to the purchase cost of the ODC. Such overhead shall collect defined costs excluded from the Labour and ISF rates.

Upon ESA request, non-audited companies shall be able to show in detail the calculation of the Labour, ISF and any procurement overhead quoted. The application of a procurement overhead shall be normal practice at the company and calculated over a normalised and optimised production level.

ODC are sourced from usual business suppliers and there are no requirements from ESA to report the actual purchase cost (except in case of audit). However, bidders are recommended to keep details of the estimated cost (for example RFQs, RFIs, price lists and the assumed quantities where relevant) in case of ESA audits, where such documentation may be requested.

Suppliers of ODC are not subcontractors (see definition below). The geographical return of ODC remains with the Country of the bidder.

Annexes

Annex A B and C provide standard tables to include a description of the planned expenditures, as a justification of the quoted total cost.

In Annex A, Materials cost shall be split among (Material type):

- Raw materials, i.e. generic materials and semi-finished products withdrawn from the company stocks;
- Parts and Products, i.e. standard/commercial products directly purchased or withdrawn from the company stocks;
- Hi-Rel / EEE Parts, i.e. high-quality products qualified with specific tests, typically used for flight equipment;
- Major sub-assemblies, i.e. parts or sub-assemblies supplied by third parties and ready to be fitted in the delivered products.

The bidder is required to indicate the quantity, whether or not the purchase is part of the Long-Lead Items list, and whether the source of supply is within or outside ESA Member States.

Annex D can be used to report the travel plan quoted.

Travel costs shall be quoted in accordance to the company's own travel regulations/policy; therefore, for example, the quoted subsistence cost shall reflect the actual cost to be incurred by the company in compensating its employees.

The Travel cost per person (p.p.) shall assume sufficient advance planning to minimise the cost and include door-to-door transport expenses (e.g. taxi, car rental, flight, train). Subsistence cost includes the cost of accommodation (e.g. Hotel), meals and personal expenses. Special compensations for work outside the duty station are allowable only when the bidder can prove that such costs are excluded from the Labour rates.

7. TOTAL COMPANY COSTS

The subtotal 4 of PBF shall give the cost of the project as expected to be incurred by the bidder. It includes Labour cost, ISF cost and ODC.

Provision for Escalation in case of FFP or CP->FFP

The PBF require that the Labour cost, ISF cost and ODC is calculated at declared economic conditions. As a consequence, the quoted rates cannot be averages across the full period of contract execution, they shall refer to a given point in time.

Therefore, in case multi-year projects, in FFP or CP->FFP, a *Provision for Escalation* can be added as a separate element and split by Labour cost, ISF cost and ODC cost, to take into account the evolution of prices within the planned contract duration and in consideration of the expenditures' distribution.

A minimum of 10% of the cost of each of the cost elements shall be deemed FIXED, therefore not subject to inflation; for the remaining 90% the bidder can propose a relevant Provision to be justified within the price proposal.

Note 1: the absence of a Provision for Escalation is acknowledged at ESA as the declared costs are inclusive of such provision, even when rates refer to specific economic conditions.

Note 2: in case of Fixed Price with price Variation (FPV) the contractor shall not include any Provision, because the price inflation is calculated at the time of the milestone payments on the basis of an agreed formula.

8. TOTAL COMPANY PRICE

The total company price is calculated by adding the Total Company Costs (subtotal 4) to the following three other elements:

Profit Fee

A profit fee (or profit margin) may be added by the bidder as a percent (%) of the subtotal 4.

The allowable % shall be no higher than the value indicated in the price requirements published with the specific Tender Action.

The maximum percentage is normally calculated in accordance with the applicable ESA Profit Policy.

In some type of contracts, e.g. co-funded, the profit fee may be considered not allowable.

Cost Without Additional Charge

Cost Without Additional Charge (CWAC) are expenses incurred on behalf of the customer and necessary for the execution of the project that are directly charged back to the customer without a markup, functioning as a reimbursement. These third-party costs, such as special support equipment, IPR license fees, services from third-parties or regulatory fees, pass through the vendor's books but are not meant to attract overheads nor generate profit.

Annex E provides a standard table to report some details of the quoted CWAC. In particular, the bidder can indicate whether the CWAC is a *Pass-Through* cost or a special *Fee*.

This type of costs includes the value of *Customer Furnished Items* (CFI) which will not be added to the total price, but it will form part of the total value of the goods delivered.

The CWAC element may be chosen for example in cases of External Services or External Major Products or Miscellaneous costs of high value with respect to the internal costs, where therefore the application of overheads and profit would be considered unreasonable by ESA.

In all cases, the purchased CWAC shall be necessary for the execution of the work. A relevant portion of the full cost shall be charged in cases where a purchased good remains property of the contractor and further used in its business.

Special cases of CWAC are:

- Management Reserves (i.e. Risk Contingencies)
- Company Contributions and co-funding

For these cases, the PBF include specific input tables.

Company Contributions

Annex F is designed to support the input of Company Contributions.

A Company Contribution is required in cases of co-funded contracts and may be a deliberate decision from bidders to implement a price reduction.

The bidder is required to define in Annex F the type of contribution:

- SELF-FUNDING, i.e. cost incurred is funded by the bidder through own reserves, sales margins and profits or acquired facilities (loans, credit).
- IN-KIND TYPE A, i.e. use of company's own resources (typically assets) excluded from the cost rates.
- IN-KIND TYPE B, i.e. value of third-party's resources available free-of-charge by the bidder.
- THIRD PARTY FUNDING, i.e. declared coverage by a third-party of part of the contract price.

In case of Contributions 'IN-KIND' (CIK), the cost or value of the contribution is expected to be part of the quoted cost (either in subtotal 4 or in CWAC), because the CIK must be used to cover a necessary expenditure for the execution of the contract's tasks; then deducted as a Company Contribution. A description of the cost covered is mandatory.

In general, the bidder is required to indicate what cost the contribution is covering (whether Labour, ISF, ODC, or CWAC). Where no specific cost element is covered, the bidder can select 'Undefined'.

A combination of different types and different coverages can be entered in Annex F.

9. SUBCONTRACTORS

A sub-contract is a contract to be entered into by the bidder with a third party for a clearly defined task, related and necessary to the proposed scope of work, and which is sufficiently non-standard to require specifications/deliverables to be procured with a relevant third-party. It is thus distinguished from a Purchase Order, which is placed on the basis of standard documents or a price list from commercial suppliers.

Subcontractors must be responsible for one or more work-packages of the WBS presented to ESA.

The cost of a Subcontractor is NOT reported within the Company Costs, neither as an ODC nor as a CWAC. A bidder is not allowed to charge any overhead nor profit on its subcontractors cost.

Subcontractors are responsible to produce their own set of PBF.

Subcontractors can themselves place sub-contracts. The structure of Prime contractor(s), level 1 subcontractors (N-1) and lower levels (N-2, N-3 ...) constitute the *Organizational Breakdown Structure* (OBS) of the industrial consortium.

A bidder (at any level of the OBS) who will place sub-contracts shall report the total price adding its own total price (i.e. sub-total 8) to the prices of its subcontractors, which are entered in annex G.

Subcontractors are classified as:

- CO-PRIMES, i.e. special case when two entities are both responsible to achieve the contract objective.
- CORE TEAM SUBCONTRACTORS, i.e. pre-selected and mandatory subcontractors part of the bidding consortium.
- OTHER SUBCONTRACTORS, i.e. any subcontractor pre-selected or to-be-selected after the contract award and not part of the bidding consortium.

10. MANAGEMENT RESERVE

Bidders or Industrial Consortiums bidding to ESA contracts can define and declare a Management Reserve to protect their affordability with risk contingencies.

Management Reserve can be calculated for the Prime and any Subcontractor in alignment to the ESA Profit Policy provisions and indications.

A specific requirement may be provided to the bidders within the Tender Action Package.

The formal Management Reserve is regulated by specific contractual conditions, which normally foresee its use only after ESA approval, and therefore the amount pre-defined becomes part of the contract price only after a Contractual Change is approved. For this reason, the Management Reserve is always included in proposals as a Firm and Fixed Amount (FFA).

The input of Management Reserve (MR) is done in Annex G for the Subcontractors MR (SMR) and in Annex E for the Baseline MR (BMR).

11. PBF-2

The PBF-2 shall give the same information, in part consolidated, as in PBF-1, but limited to each work-package, which is represented by separate columns.

Every sub-total amount in the last column (sum of all previous columns) shall be equivalent to the sub-totals in PBF-1, both in terms of NC and in EURO.

The bidder is expected to report every work-package as defined in the WBS presented to in the proposal.

When compiling manually the MS Excel version, the bidder shall add columns as necessary.

12. PBF-3

The PBF-3 shall give the same information, in part consolidated, as in PBF-1, but limited to each time period. The standard time period represented in columns is 'month', but for long-durations projects it may be extended to 'quarter' (i.e. 3 months totals) or even 'year' totals. Every sub-total amount in the last column (sum of all previous columns) shall be equivalent to the sub-totals in PBF-1, both in terms of NC and in EURO.

The bidder is expected to report a realistic expenditure profile, therefore representing

- Realistic personnel usage profiles (i.e. hours distribution over time);
- Realistic use of ISF;
- Realistic payments to suppliers and LLIs.

In the same way, the subcontractor's payment plans shall be reported as precisely as possible. When such information is not available, the bidder is expected to give a best estimate of that.

Note: bidders shall refrain from reporting costs as average distributed values from start to the end of the work package (unless this profile is truly realistic) because by doing so they invalidate the profile of expenditures and the verification of the fair payment plan.

When compiling manually the MS Excel version, the bidder shall add columns as necessary.

Payments Plan justification

The bottom lines of the table are dedicated to the milestones' payments.

The bidder is required to report the proposed payment plan according to the contractual dates. Any *Advance Payment* requested by the bidder shall be included in the first column and deducted from the milestones amounts as applicable. This approach will project the expected cash flow.

The difference (P-E) between cumulative payments (P) and cumulative expenditures (E) gives the deviation in cash balance at a given time. A graph (similarly to the previous PSS-A15.1) can be realised using the last two lines of the form.

13. SUPPORT

Support can be provided to bidders as follows:

- A. For question regarding specific Tender Actions, please contact the ESA Contract Officer indicated in the Tender Action Package.
- B. For general questions regarding rates structures, rates audits and cost allowability, please contact the ESA Industrial Audit Team via pss@esa.int or esa-star ASTRA services.
- C. For questions related to ECOS, please contact esait.service.desk@esa.int

14. NOTE REGARDING THE PSS-A FORMS

The PSS-A forms are standard tables derived from the old space standards for Project Management and Cost Control.

The PBF replace the issue 5 of the PSS-A forms, maintaining as much as possible the structure and nomenclature and introducing necessary changes.

The PBF represent the issue 6 of the ESA standard price reporting tables.